

Banking Information System

Summer Internship Final Project Report

Intern Name Pradeep

Organization / Platform upSkill Campus / UniConverge Technologies (UCT)

Project Domain Java Software Development & Object-Oriented Programming

Repository Name upskillcampus

1. Executive Summary

This project presents an object-oriented Banking Information System developed in Java. The system simulates core banking operations, including account creation, balance inquiry, deposit management, and withdrawal verification. Built using robust Encapsulation principles, it ensures strict data privacy and validates account transactions in real time.

2. System Architecture & Features

Account Management: Securely stores account details (Account Holder Name, Account Number, Current Balance) using encapsulated object attributes.

Transaction Logic: Implements conditional validation checks for deposits and withdrawals, preventing invalid transactions or negative balance errors.

Console Telemetry Interface: Displays structured transaction receipts and updated account details directly to the user interface.

3. Technology Stack

Programming Language: Java (JDK 8 or higher)

Development Tools: VS Code / Eclipse IDE / Replit

Version Control: Git & GitHub

4. Implementation Source Code

```
import java.util.Scanner;
public class BankingInformationSystem {
    private String accountHolder;
    private String accountNumber;
    private double balance;

    public BankingInformationSystem(String accountHolder, String accountNumber, double initialBalance) {
        this.accountHolder = accountHolder;
        this.accountNumber = accountNumber;
        this.balance = initialBalance;
    }

    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Successfully deposited: $" + amount);
        } else {
            System.out.println("Invalid deposit amount.");
        }
    }

    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Successfully withdrew: $" + amount);
        } else {
            System.out.println("Insufficient balance or invalid amount.");
        }
    }

    public void displayDetails() {
        System.out.println("\n--- Account Details ---");
        System.out.println("Holder Name: " + accountHolder);
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Current Balance: $" + balance);
    }
}
```

```
public static void main(String[] args) {  
    BankingInformationSystem account = new BankingInformationSystem("Pradeep", "ACC100293", 1000.00);  
    account.displayDetails();  
    account.deposit(500.00);  
    account.withdraw(200.00);  
    account.displayDetails();  
}  
}
```

5. Project Links & Verification

GitHub Repository: <https://github.com/janani18730/upskillcampus>

Code File Link: <https://github.com/janani18730/upskillcampus/blob/main/BankingInformationSystem.java>

Report PDF Link:

https://github.com/janani18730/upskillcampus/blob/main/BankingInformationSystem_Pradeep_USC_UCT.pdf

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